

Polynomial Regression

Final methodology and validation results | Student BT2024146

1. Problem and data

The assignment contains two separate supervised regression problems. Each model predicts the continuous target **y** from its assigned input features. Both datasets contain 1,000 labeled training rows and 1,000 test rows. **var1** has six features and permits polynomial degrees 1 to 10; **var2** has three features and permits degrees 1 to 20. Test labels are unavailable, so no test-set error is reported.

2. Polynomial-regression methodology

PolynomialFeatures expands the inputs to include monomials up to the candidate total degree. Ridge regression is evaluated across the full permitted degree range, with its penalty strength selected from the candidate values in the model-selection code. Ordinary least-squares polynomial regression is also compared when the expanded design contains fewer than 800 terms. The scaler is fitted separately on each fold's training partition and then applied to that fold's validation partition, preventing preprocessing leakage.

3. Validation and selection

Validation setting	Configuration
Strategy	Repeated 5-fold cross-validation; 2 repeats (10 folds)
Randomization	Shuffled folds; random seed 2026
Rows per fold	800 training rows and 200 validation rows
Selection	Lowest degree within one standard error of the minimum mean validation MSE

Final selected models

Problem	Model	Degree	Alpha	CV MSE	CV R-squared
var1	Ridge polynomial	5	20	0.515157	0.951535
var2	Ridge polynomial	10	1	0.262251	0.994186

Degree 11 achieved the lowest raw cross-validation MSE. However, degree 10 was selected using the one-standard-error rule because its validation error was within one standard error of the minimum while requiring fewer polynomial terms.

For var2, the raw minimum was 0.260706 at degree 11; the selected degree 10 result was 0.262251, below the one-standard-error cutoff of 0.268001. Degree 10 uses 285 polynomial terms, compared with 363 at degree 11.

The training-fold errors are lower than the validation errors for both selected models, showing a training-to-validation gap. Selection is therefore based on held-out fold performance and a lower-complexity degree preference, rather than training fit alone.

4. Final fitting and prediction files

After model selection, each selected polynomial Ridge pipeline is refitted using all 1,000 labeled rows for its problem. It then predicts the matching test dataset, preserving that file's row order. The two problems are trained and predicted independently.

File	Verification
BT2024146_pred_var1.csv	1,000 finite predictions; one column named y; no index column
BT2024146_pred_var2.csv	1,000 finite predictions; one column named y; no index column

5. Input checks

The training and test feature schemas match for both problems. The supplied data contain no missing values. Across the supplied feature columns, values range from -1.000000 to 1.000000. The prediction files were checked for their required single-column schema, 1,000-row count, and finite numeric values.

6. Reproducibility

Place the five instructor-provided CSV files in the project's **data/** directory: the train and test files for var1 and var2, plus sample_submission.csv. These inputs and the assignment PDF are not required repository deliverables. Install the pinned packages from requirements.txt, then run the following from the repository root:

```
python src/train_and_predict.py
python src/build_report.py
```

The first command runs degree and model selection, refits the selected models, writes the prediction CSVs, and saves validation and data-check results under outputs/. The second builds this report from those result files. See README.md for the complete Windows PowerShell environment setup and package installation commands.

Code locations: src/model_selection.py contains the fold-safe model search; src/train_and_predict.py performs data checks, final training, inference, and prediction-file validation; src/build_report.py generates this PDF from the saved results.